Teach a machine to say “I don't know”

Contracts and forms are typed into SIXT's systems by hand. AI can do it in seconds — but can we trust it?

1 human review
costs 1

1 wrong value booked
costs 100

→ automation only pays off
if booked values are right
> 99% of the time

Where we started: trust the model's confidence

One LLM call per document; book everything with confidence ≥ 0.9.
Result on 20 held-out contracts:

52 booked · 12 wrong
net saving −140%

Worse than typing everything by hand:
12 confident mistakes cost as much as 1,200 reviews.

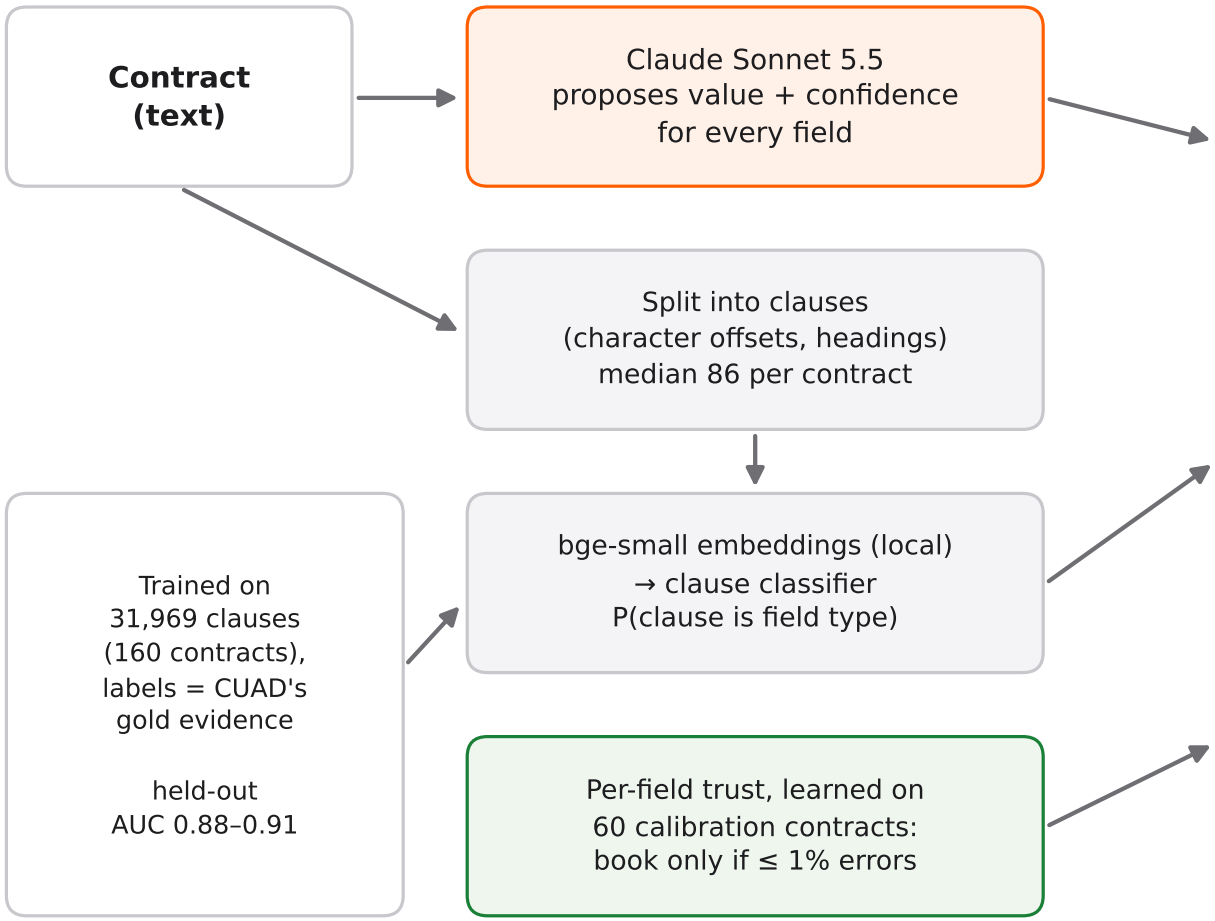
Why this is hard (measured in our EDA)

- 68% of contract fields are empty → a model that always answers invents values
- 28% of contracts are longer than the prompt limit
- 45% of form PDFs are scans with no text to check against
- Keyword search finds the right clause for only ~58% of fields
- The model's confidence is not calibrated: right and wrong answers both come back at 0.9–0.95

The real problem is not extraction.
It is knowing when you are right.

Contracts: the model proposes, evidence decides

Method FINAL (CUAD, 41 clause fields) · 1 LLM call per contract · declared threshold 0.95



Decision rules — first match wins, nothing averaged

1. model returned null	ABSTAIN
2. value not in the contract	REJECT
3. clause is another type (p < 0.02)	REJECT
4. clause type uncertain (p < 0.10)	REVIEW
5. other clauses of this type left out	REVIEW
6. fragment, not whole sentences	REVIEW
7. field not trusted on calibration data	REVIEW
8. model confidence < 0.95	REVIEW
9. every check passed	BOOK

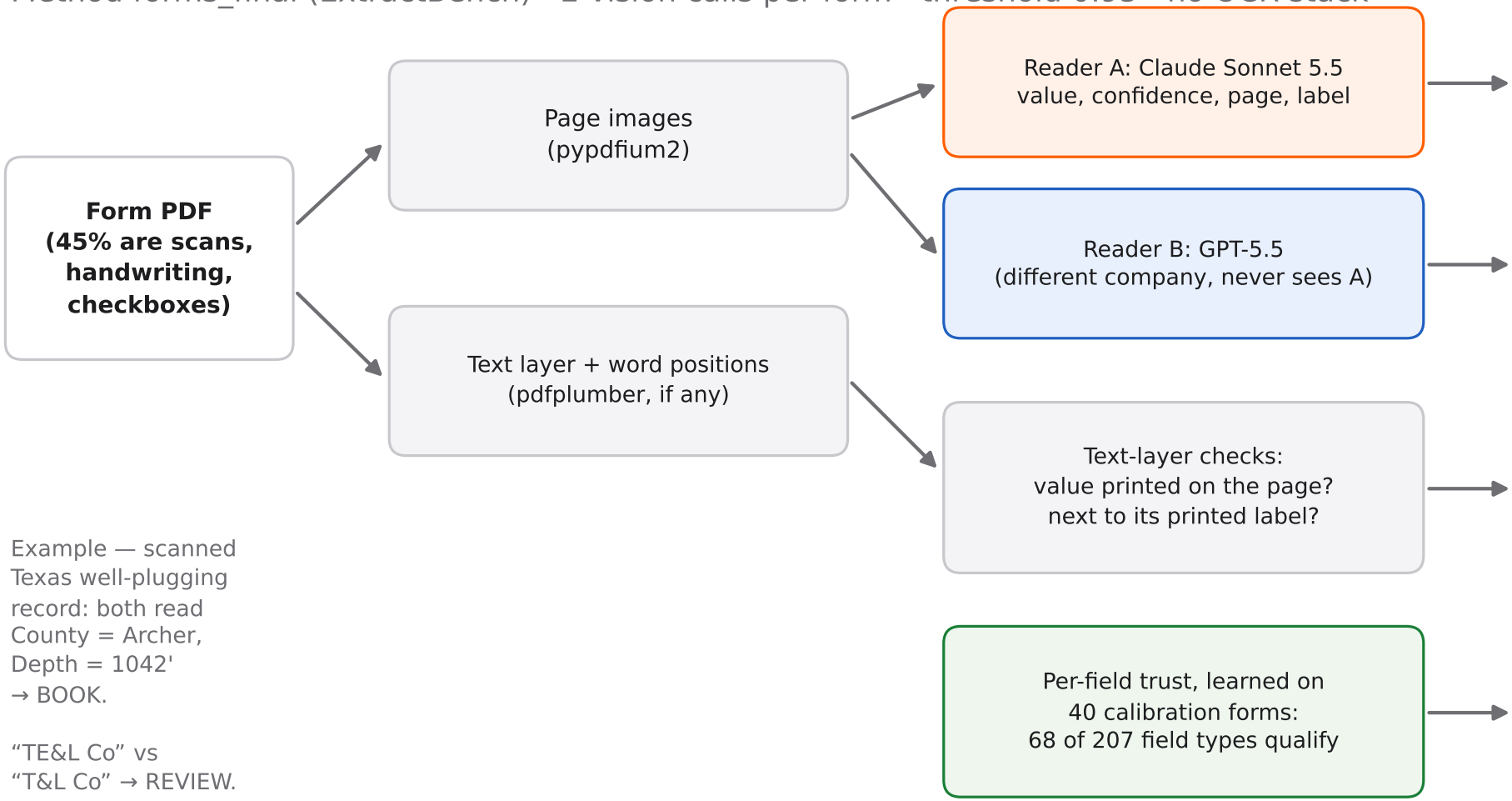
Confidence is the last check, never the main one.

Example — Non-Disparagement: the model quoted a real sentence about a domain licence; its clause scores p = 0.0005 for that type → REJECT (field truly absent).

If the classifier cannot load, every contract field goes to review (fail-safe).

Forms and scans: two independent readers must agree

Method forms_final (ExtractBench) · 2 vision calls per form · threshold 0.95 · no OCR stack



Example — scanned Texas well-plugging record: both read County = Archer, Depth = 1042' → BOOK.

“TE&L Co” vs “T&L Co” → REVIEW.

Decision rules

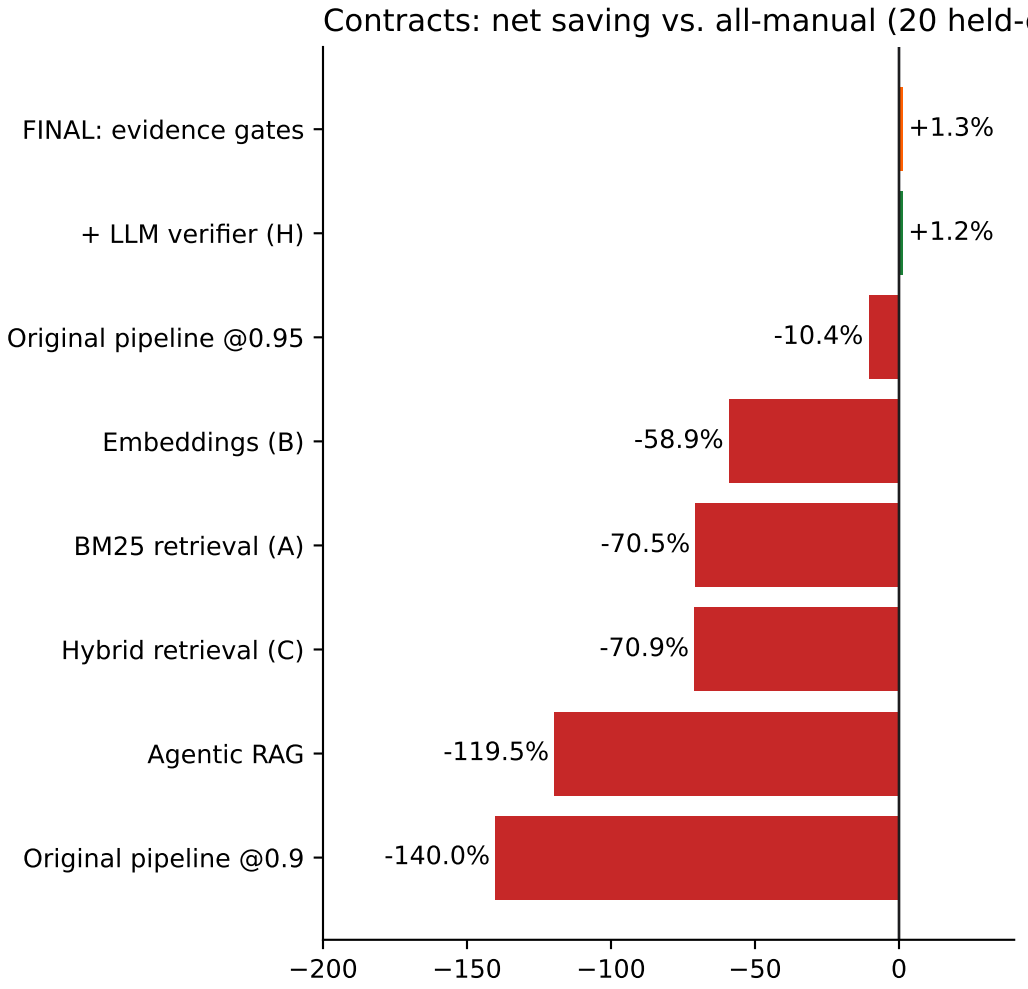
1. both readers: nothing	ABSTAIN
2. one reader only / disagree	REVIEW
3. not printed on the page	REJECT
4. not next to its label	REVIEW
5. field not trusted	REVIEW
6. confidence < 0.95	REVIEW
7. all passed	BOOK

Why two model families?

A scan has no text to check against. Two independently trained models rarely make the same mistake: 39 of 46 disagreements were real reading errors.

Results: from −140% to 98 of 99 bookings right

Official scorer, declared threshold 0.95 · held-out and untouched data only



Untouched public test (run_hidden.sh, gold removed)

	booked	match gold	net saving
Contracts (24)	11	11	+1.1%
Forms (16)	88	87*	−1.4%
All 40 docs	**99**	**98***	**−0.1%**

*** The one mismatch is a gold-label error**

The form prints “3583 CASTLEHILL WAY”; both readers copied it exactly;
the gold says “CASTLE HILL” (checked in the text layer, the image and by hand).
Counted by the scorer → break-even. Corrected → 99 / 99, +5.4%.

Held-out dev data (52 docs, 2,384 fields): 92 / 92 booked values correct,
0 hallucinations, net saving +3.9% — the same at an error cost of 20, 100 or 500.

Forms alone (32 held-out): 81 / 81 correct, +5.2%.
Without the field-trust rule: 3 wrong, −1.8%.

What we really learned about the problem

Every point below was measured on held-out data

Confidence is not evidence.

Raising the threshold from 0.9 to 0.95 even lowered one method's precision (71.7% → 68.4%). Independent checks decide; confidence is the last one.

The main error was incomplete, not wrong.

CUAD's gold lists every passage (2–5) for a clause; the model quotes one. The completeness rule catches these.

More LLM passes did not help.

Three verifier models and an agent added cost; better retrieval raised raw accuracy (77% → 80%) but not booking safety.

Disagreement is a strong signal.

On forms, 39 of 46 disagreements between two model families were real errors. But agreement alone still let 3 wrong values through.

Learned per-field trust makes booking safe — and is now the bottleneck.

Scoring the evidence sentence fixed 8–20 false “uncertain” verdicts, yet bookings stayed 11 → 11: those fields lack calibration evidence.

Gold labels have errors too.

At a 99% target, one label error decides break-even vs. +5%.

Next steps

- Calibrate on all 312 / 127 training documents → more trusted fields
- Ask for all passages of a clause (the main contract error)
- Normalise number formats in the forms check (43 correct values lost)
- Use reader B when reader A misses a value (229 cases)

How SIXT could roll it out

Shadow mode first: humans see suggestions and reasons; switch on booking field by field once its measured error rate < 1 / error cost.

Demo: upload a document → every field's decision, reason and a citation box in the original PDF (Project_1/demo_app.py).